Technical details: compile check

Technical details

Pre-processing. Every real exposure was converted to attenuation (the negative logarithm of a linear detector signal against its air or flood value, or a linear reading of a stored logarithmic display), then windowed identically for every device: the field's 1st, 5th, 50th, 95th and 99th percentiles were mapped to grey levels 27, 47, 126, 216 and 239, bone bright, and stored as an 8-bit image. A truth box is the projection of every voxel of a cone-beam CT vertebra label (subsampled to at most 20,000 points) through the registered pose, taking the 2nd–98th percentile extent; a box was kept when at least half its projected points fell inside the image and both of its sides measured at least 8 pixels. Level crops are fixed-scale two-channel windows of 320 pixels (image and target marker) with the target body at 96 pixels.

Acquisition. Table S1 lists the acquisition parameters each release carries. No release documents an exposure protocol beyond its headers, and no parameter was controlled by this study.

Models, initialisation and training. The sacrum detector is YOLO11m (Ultralytics; classes vertebra and sacrum; 1,024-pixel input), initialised from the COCO-pretrained YOLO11m weights through a render-trained checkpoint and trained for 80 epochs (patience 25, batch 8, seed 0, deterministic) with rotation up to 10°, translation 0.1, scale 0.3, horizontal flip 0.5 and mosaic 0.5, on the training and validation frames of COBRA2026 and DeepFluoro; its checkpoint was selected on their test frames, so only its held-out-device results are clean. The body detector is YOLO11m-pose fine-tuned for 60 epochs from an earlier pose checkpoint on the plain-film keypoint corpus, used at confidence 0.35 **[PENDING G5: keypoint detector: training corpus by source and split, and whether any film of a test person is in it]**. The level encoder is ResNet-18 with a two-channel stem and a 27-level head (11,173,376 parameters; 3.62 giga multiply–accumulates per crop). The cross-validation encoders were initialised from a level en-

coder trained on renders and films (registry model allspine_v2_r1) and fine-tuned per fold for 15 epochs at a learning rate of 10^{-4} with AdamW (weight decay 10^{-4}), a cosine schedule and class-weighted cross-entropy. The encoder whose embedding Figure 2 maps is the allspine_v9 model (SHA-256 188f9fcb…). The recurrent head (linear 512 to 256, GELU, two-layer bidirectional GRU of width 256, linear 512 to 27; 2,117,403 parameters) was trained on runs of two to nine consecutive embeddings of mixed length for 30 epochs at a learning rate of 10^{-3} (AdamW, weight decay 10^{-4}, cosine schedule) with per-position cross-entropy. No ensemble is used; each crop is also read horizontally flipped and the two readings averaged. Counting and the alphabet coupling have no learned parameters. The configuration scored in Figure 6 is **[PENDING G5: the G5 manifest: bank, split, initialisation and recipe of the arm never trained on a test person]**.

Table S1: Acquisition parameters of the real exposures, as released.

Collection	Device	Detector pitch (mm)	Frame (pixels)	Tube voltage, kVp (5th–95th percentile)
C-ArmLSpine	Surgio B51 flat-panel C-arm	0.15 (a)	2,048 × 2,048; exported at 1,024	73–111
DeepFluoro	Siemens Cios Fusion	0.194	1,536 × 1,536	not recorded (b)
COBRA2026	Elekta XVI kV imager	0.8	504 × 504	120
Geneva hip	Philips BV Pulsera image intensifier	0.2904	1,000 × 1,000	65–90
Ljubljana 2D-3D-GS	Trixell PIXIUM 4600 flat panel	0.286	1,560 × 1,560	not recorded

(a) 307.2 mm over 2,048 pixels; the header pixel spacing is a placeholder of 1.0. Source-to-image distance 1,140 mm, source-to-object distance 745 mm, object-plane pixel 0.1915 mm at the 1,024-pixel export. (b) Present in the vendor DICOM headers of the same acquisitions but not joined per registered frame.